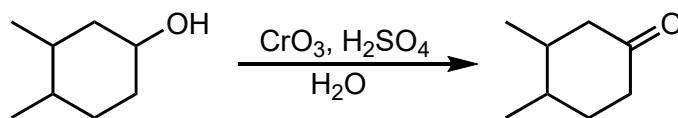
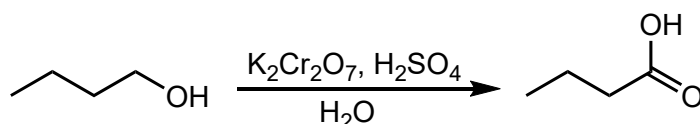


Recalling What We Already Know

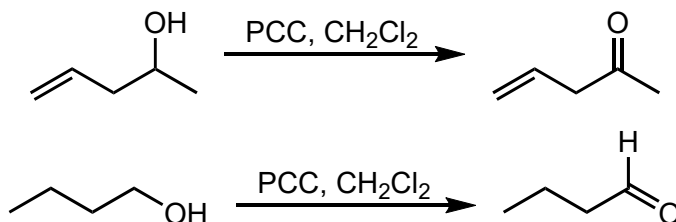
Oxidation of Alcohols



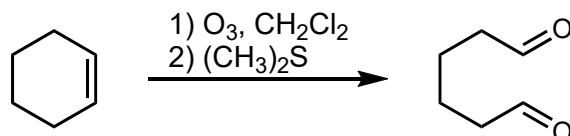
But given the presence of water with these conditions, primary alcohols are oxidized to carboxylic acids. This is due to the oxidation of the 1,1-diol (hydrate) that is produced under these conditions. Its formation is covered in this chapter.



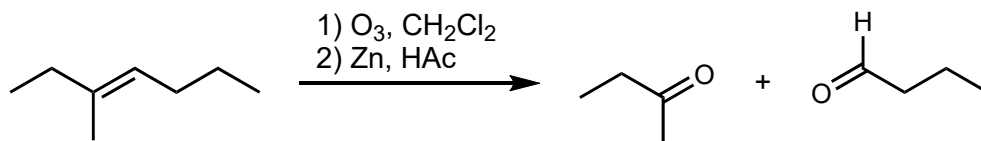
We can use anhydrous conditions (note the inclusion of CH_2Cl_2 as solvent) to oxidize in a similar fashion, but importantly, prepare aldehydes from primary alcohols.



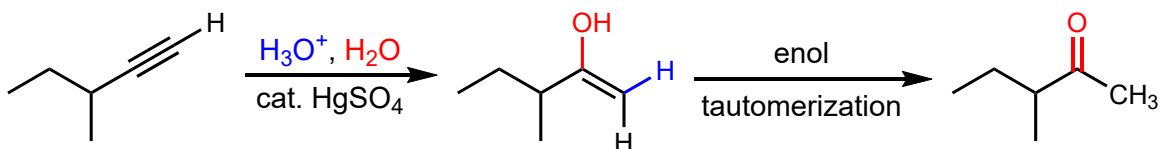
Ozonolysis of Alkenes



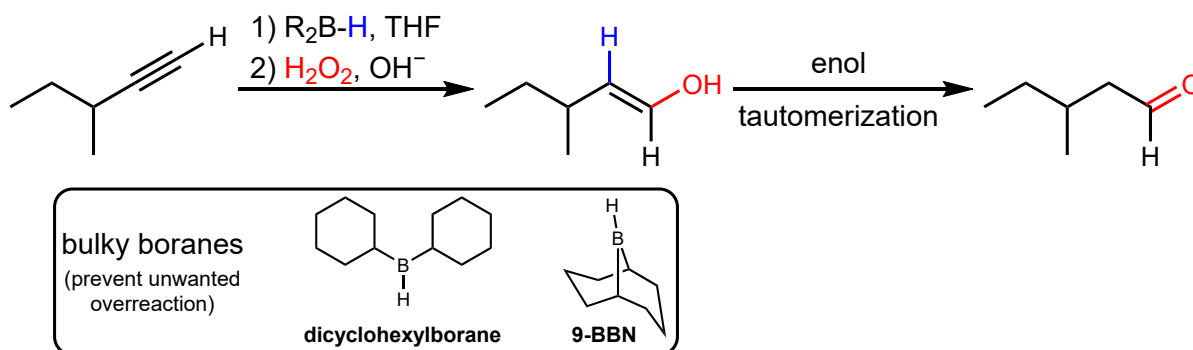
Ozonolysis allows the conversion of alkenes to carbonyls as either ketones or aldehydes. After treatment with ozone, O_3 , the addition of a reducing agent such as dimethyl sulfide, $(\text{CH}_3)_2\text{S}$, or zinc will yield the product.



Hydration of Alkynes

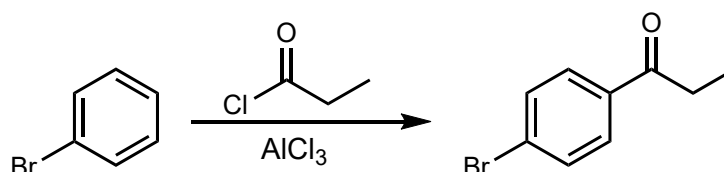


Alkynes produce enols when hydrated. But remember! Enols tautomerize (swap a proton from the alcohol to the nearby carbon) to produce the more thermodynamically stable ketone or aldehyde. Terminal alkynes will give methyl ketones when treated with a Hg^{2+} salt.



Hydroboration-oxidation of alkynes yields the anti-Markovnikov enol product which also will consequently tautomerize. In the case of a terminal alkyne this gives an aldehyde. A bulky borane is used to prevent a second hydroboration of the alkenyl borane intermediate. Check the introduction of this reaction earlier to refresh your memory, but it is sufficient to recall these conditions and their outcome.

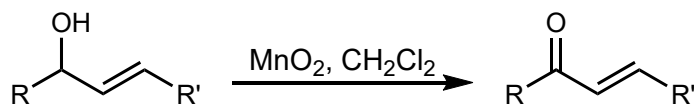
Friedel-Crafts Acylation



Aryl ketones can be synthesized by Friedel-Crafts acylation of an activated (or halogenated) aromatic compound (look for an *ortho/para*-director).

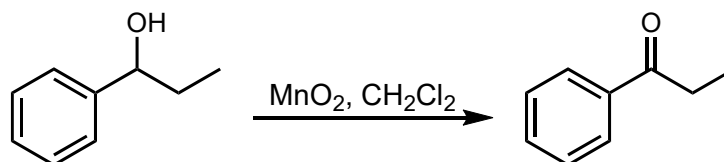
A New Option Following Recent Topics with Allylic and Aromatic Alcohols

Allylic Oxidation



An alcohol in the allylic position is subject to oxidation using MnO_2 . The α, β -unsaturated carbonyl compound will be the focus of reactions in a later chapter, but until then we can practice preparing them.

Benzylic Oxidation



A benzylic alcohol – the benzylic position is the $-\text{CH}_2$ attached immediately to an aromatic ring – will also be oxidized in the same way as an allylic alcohol.

In both these cases, be on the lookout for substitution at that position. You can't oxidize away a C-C bond with these conditions so no tertiary alcohols.